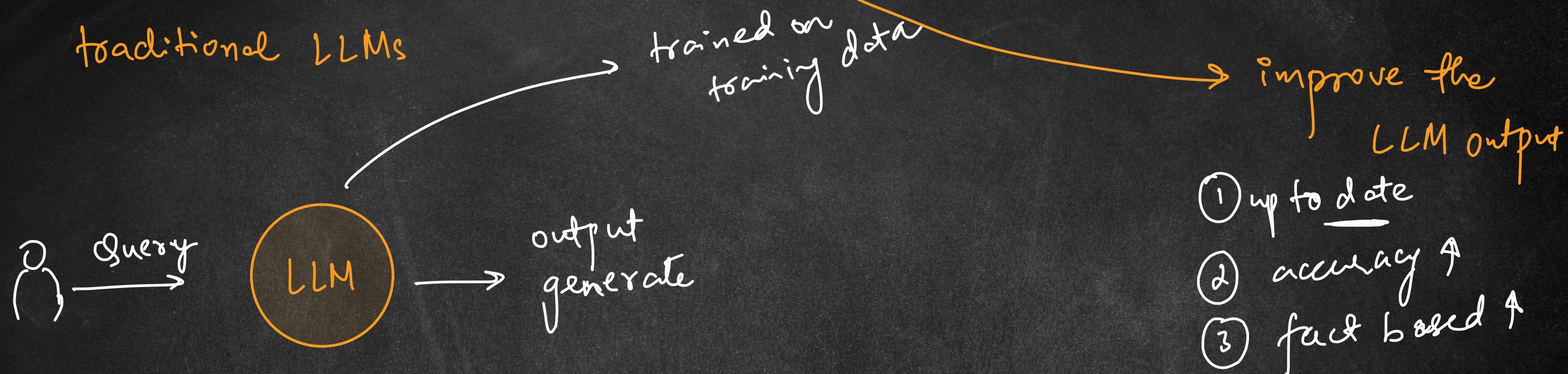


Retrieval-Augmented Generation (RAG) → Perplexity AI

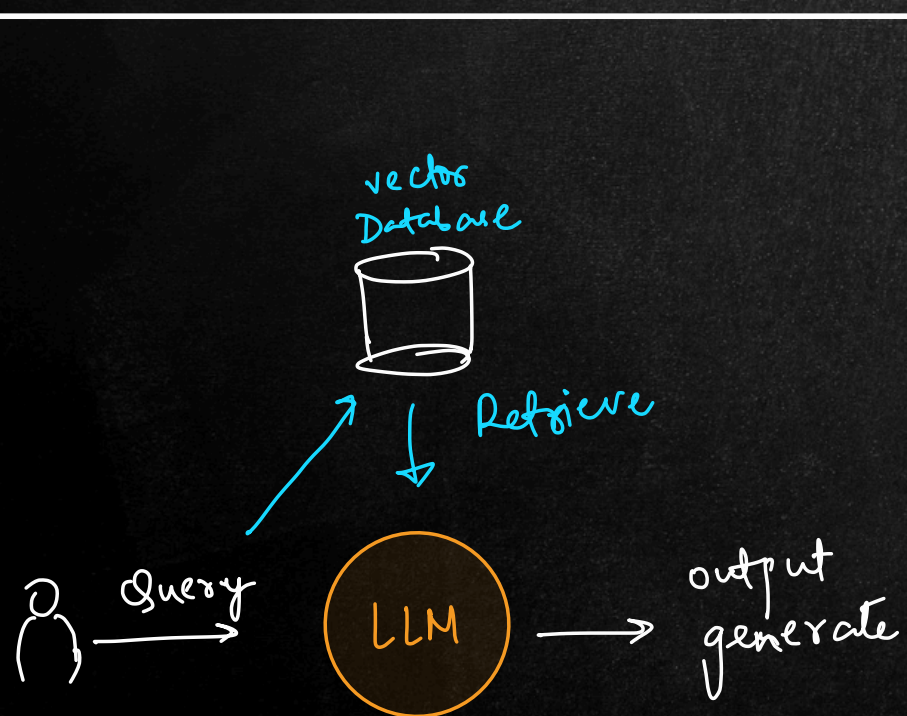
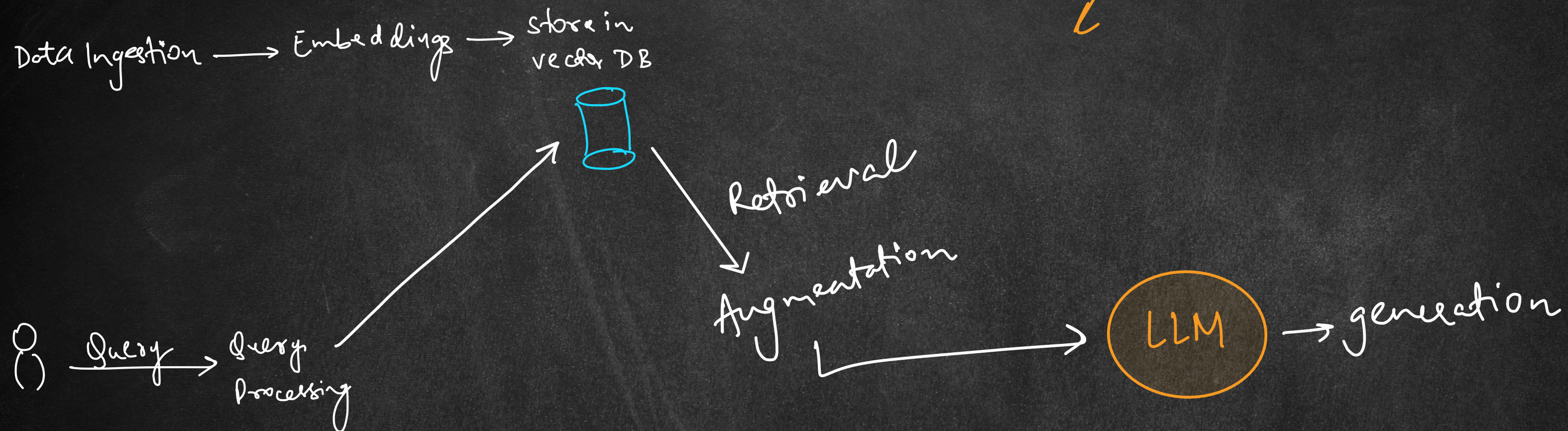


- ① Hallucination — making things up
- ② Not up to date

Information Retrieval

RAG

Architecture



SQL - traditional DB
based
name = "Harry Potter"

vector DB
→ wizards,
children fantasy

RAG

Data Ingestion Pipeline

context size

LLM

text $\rightarrow$ nums

vector embeddings

① Data Ingestion

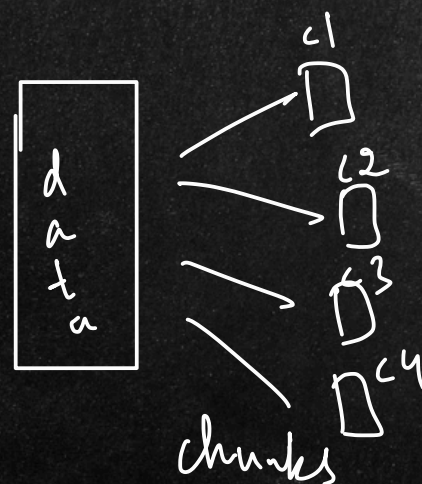
pdf
html
db
excel
...

process/clean
 $\downarrow$
chunks

② Embedding

- OpenAI embedding model

- sentence transformer



③ Vector DB

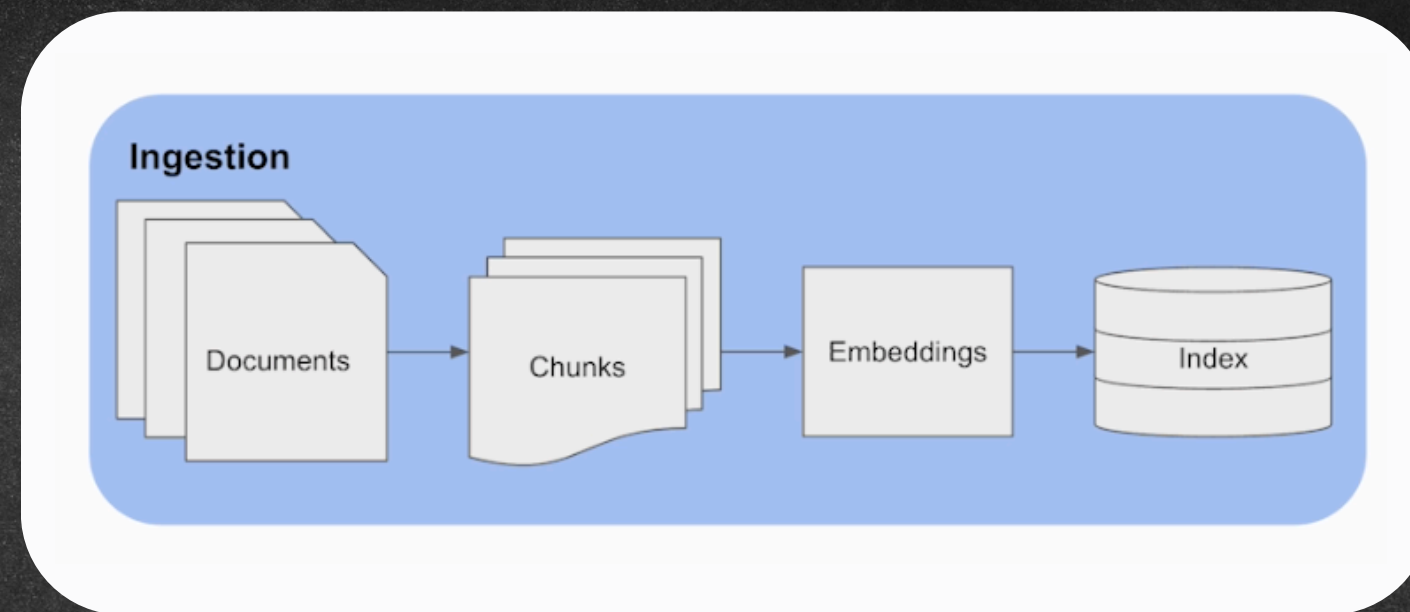


Pinecone
FAISS

chromadb

...

Knowledge Base

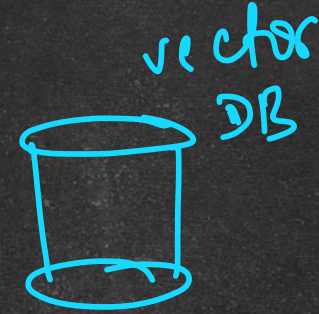


RAG

Retrieval Pipeline

semantic / similarity search
Retrieval

Query
④ processing
Query Embeddings

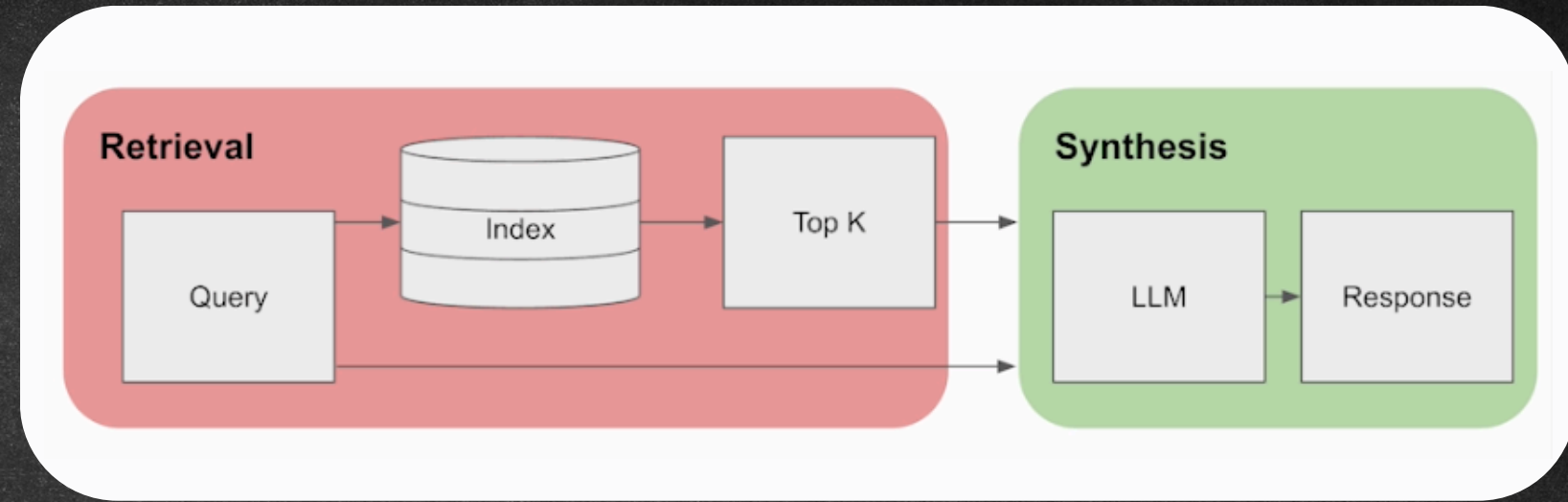


⑥ Augmentation
top K similar results
+ prompt
context



o/p generate

⑦ Generation



- Nearest neighbour search

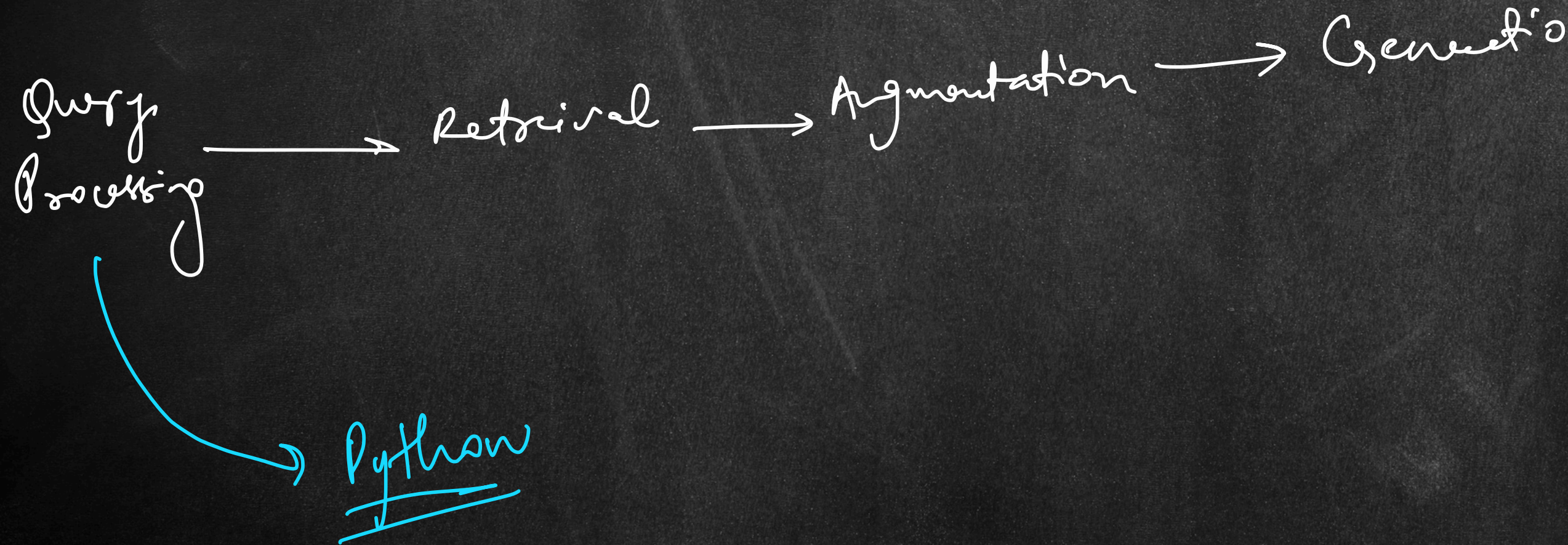
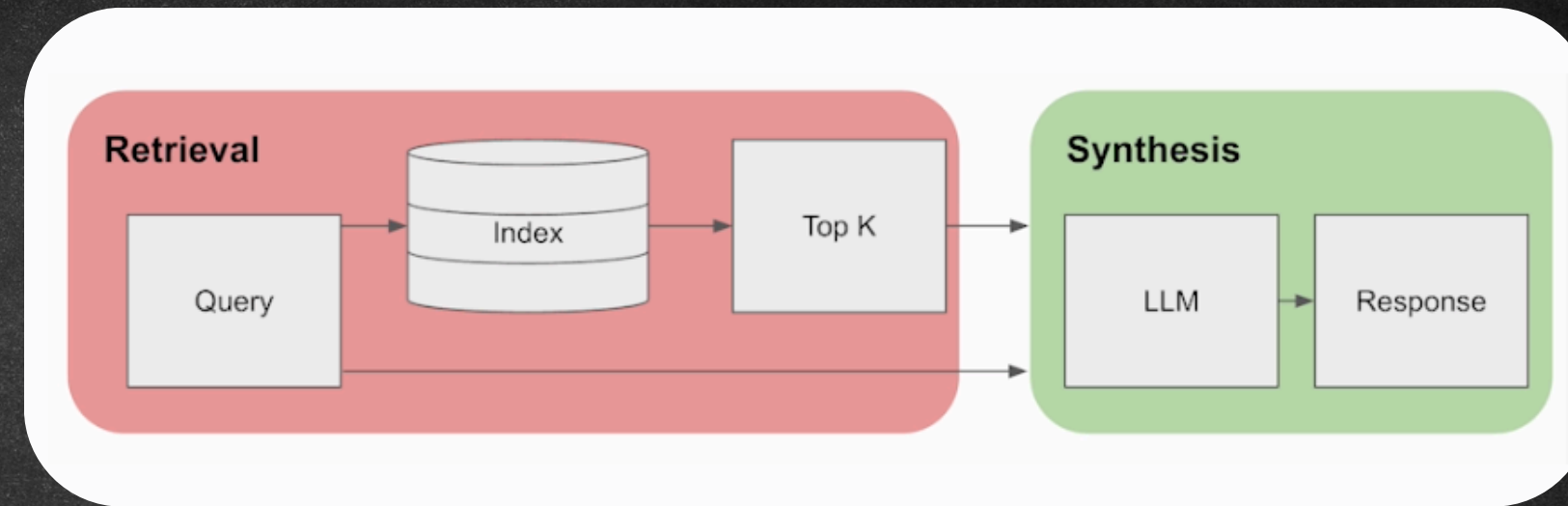
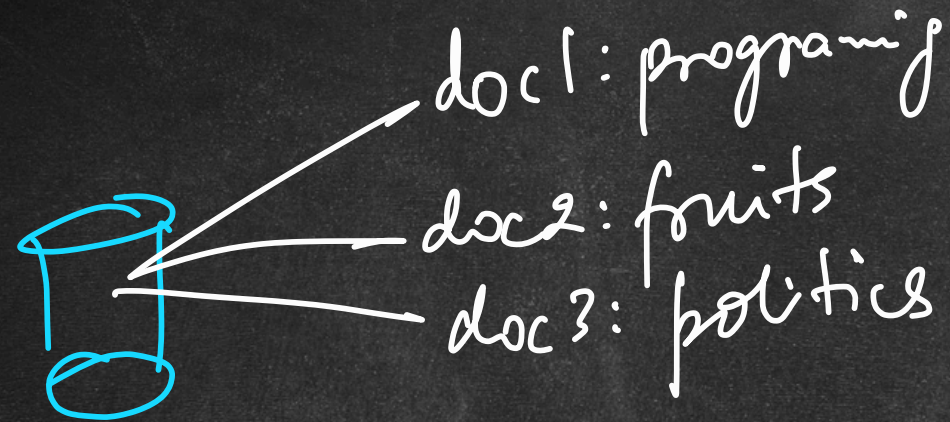
- Cosine Similarity (vectors) ↑

cosine dist ↓

$$\text{cosine similarity} = 1 - \text{cosine dist}$$

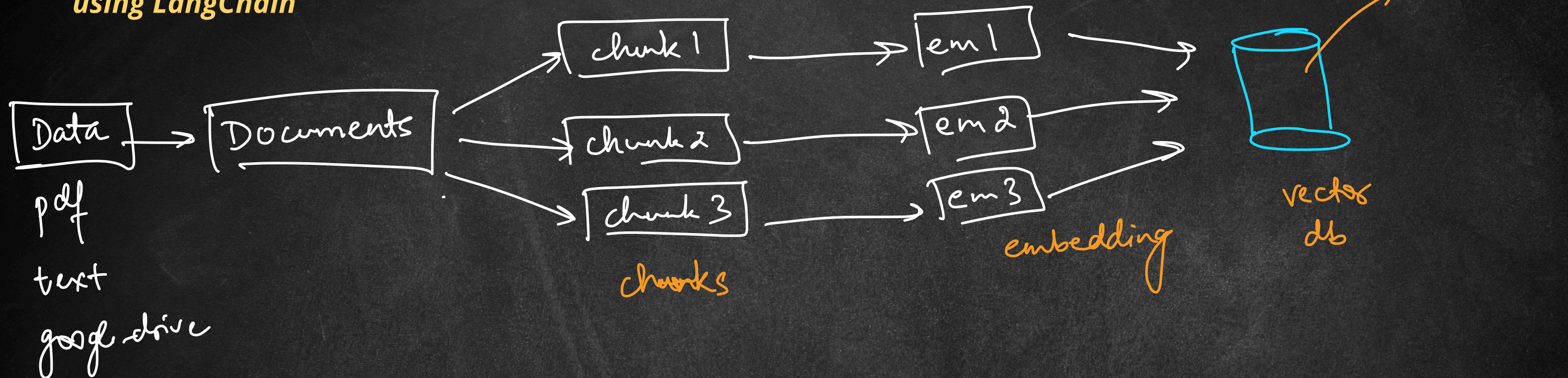
RAG

Retrieval Pipeline



Ingestion Pipeline Implementaion

using LangChain



① Document → object

{ ① text / page-content

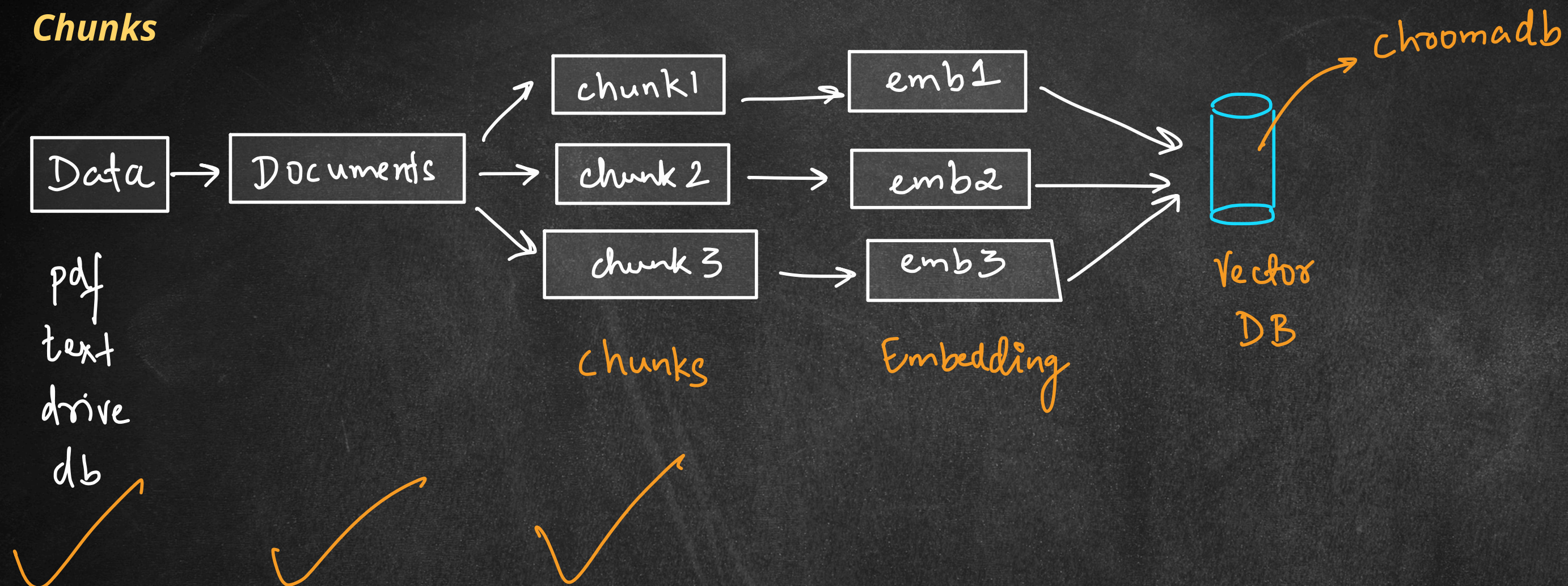
② metadata

}

② Document loaders

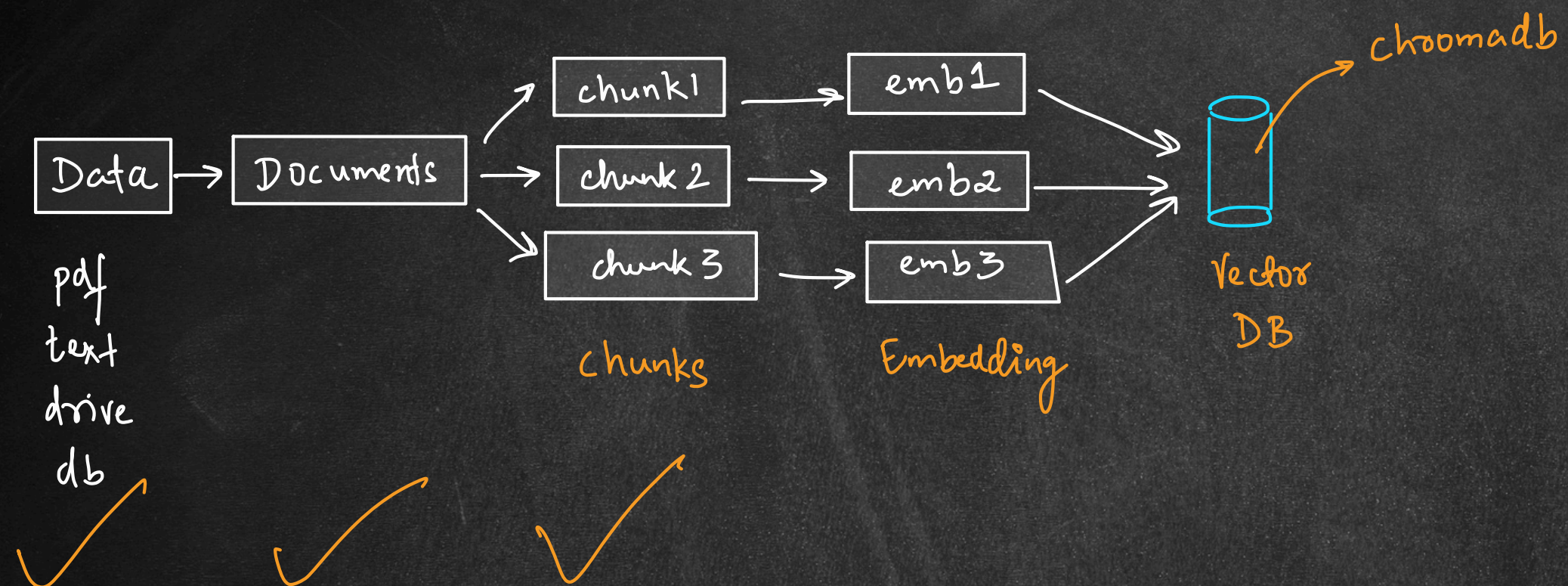
Ingestion Pipeline Implementaion

Chunks



Ingestion Pipeline Implementaion

Embedding



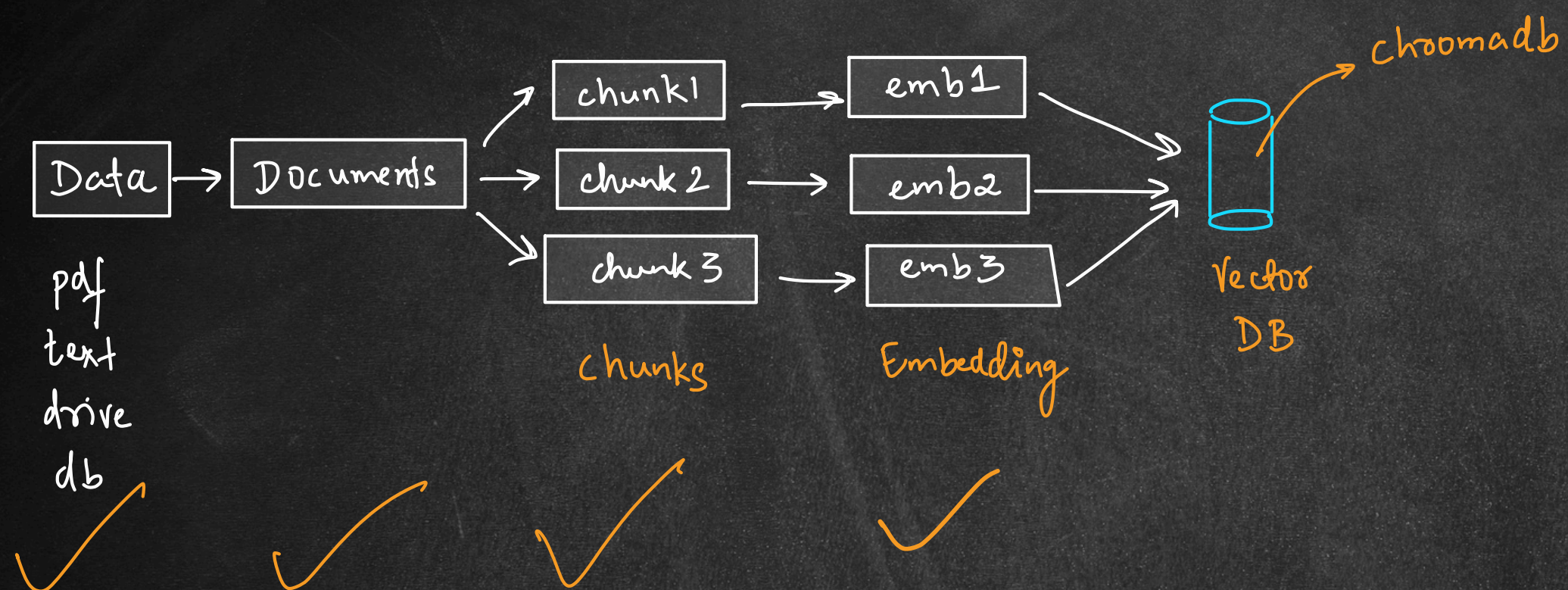
Embedding Manager

obj
code reusability ↑

Query Embedding

Ingestion Pipeline Implementaion

Vector Store

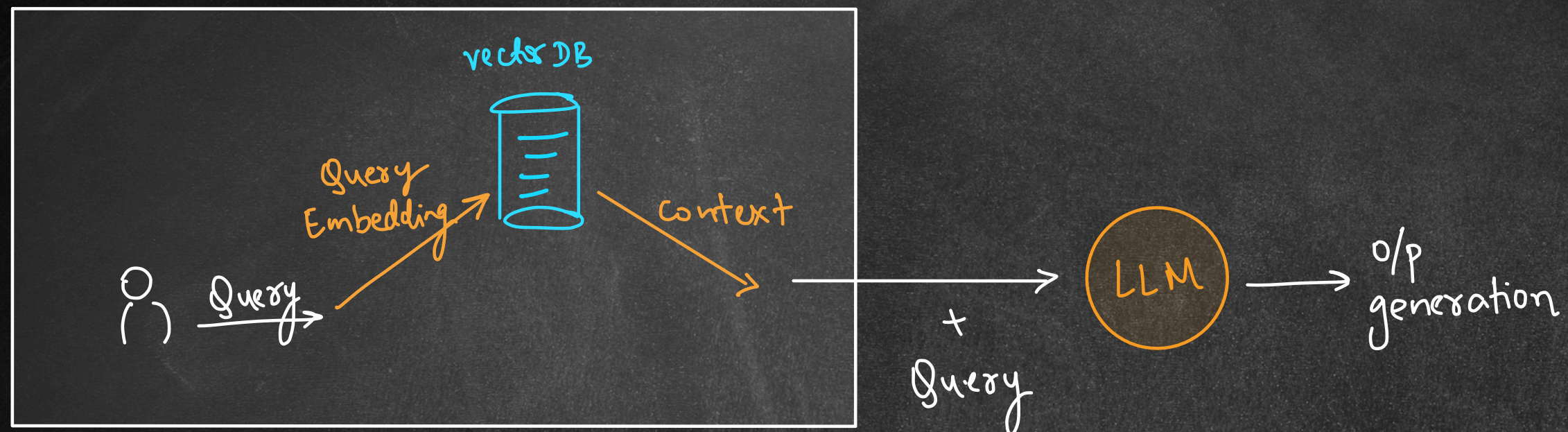


Vector Store Manager

- ① initialize vector store - connect
- ② vs_initialize → path, collection
- ③ store docs → VS

Retrieval Pipeline Implementaion

Retriever



RAG Retrieval

↓
return a list of results
(context)